Architect’s Brief

Smart Building Energy Management System

1. Architecture Purpose

This brief prepares the Smart Building Energy Management System for system architecture modeling. It identifies the likely system boundary, external participants, internal responsibilities, integrations, data ownership concerns, workflows, and quality drivers.

This brief is not a product requirements document and does not define implementation design or code structure.

2. System Boundary

The subject system is the **Smart Building Energy Management System**.

The system owns telemetry ingestion, telemetry normalization, monitoring, alerting, recommendation generation, control-action coordination, reporting, audit history, and integration status tracking.

The system does not own low-level equipment control firmware, the building automation system itself, physical sensors, physical meters, utility pricing sources, weather data sources, life-safety systems, or external notification infrastructure.

3. External Participants

Participant	Role in the Architecture
Facilities Manager	Reviews building energy performance, approves recommendations, manages policy, and monitors outcomes.
Building Operator	Responds to alerts, applies operational changes, and overrides actions when needed.
Sustainability Analyst	Reviews trends, savings, carbon estimates, and portfolio performance.
Occupant	Provides comfort feedback or environmental issue reports where enabled.
System Administrator	Configures buildings, integrations, users, thresholds, and automation policies.
Building Automation System	Provides equipment state and executes approved control commands.
Energy Meter	Provides energy usage telemetry.

Participant	Role in the Architecture
Environmental Sensor	Provides environmental and occupancy telemetry.
Utility Pricing Feed	Provides pricing and demand-response inputs.
Weather Data Provider	Provides weather observations and forecasts.
Notification Provider	Sends alert and operational notifications.
Identity Provider	Provides authentication and role information where configured.

4. Candidate Internal System Responsibilities

Responsibility Area	Description
Building Configuration Management	Manages buildings, floors, zones, meters, sensors, equipment groups, schedules, and thresholds.
Telemetry Ingestion	Receives readings from meters, sensors, gateways, and building automation systems.
Telemetry Validation	Detects invalid, missing, stale, duplicated, or out-of-range readings.
Telemetry Normalization	Converts heterogeneous incoming readings into a common internal representation.
Time-Series Management	Stores and retrieves historical telemetry for dashboards, analytics, and reporting.
Real-Time Monitoring	Provides current energy, environmental, equipment, and freshness views.
Anomaly Detection	Detects abnormal energy usage, comfort issues, stale telemetry, equipment runtime issues, and integration failures.
Recommendation Generation	Produces energy-saving or cost-saving recommendations with rationale and constraints.
Approval Management	Captures approval, rejection, override, and dismissal decisions.
Control Coordination	Sends approved control commands to the building automation system and tracks outcomes.
Demand-Response Coordination	Evaluates events, proposes actions, tracks

Responsibility Area	Description
	participation, and summarizes results.
Weather and Pricing Context Management	Ingests and applies external weather and pricing data to decisions.
Alert Management	Creates, routes, updates, acknowledges, and closes alerts.
Notification Management	Sends alert, reminder, escalation, and operational notifications.
Reporting and Analytics	Produces usage, cost, savings, runtime, carbon, and portfolio reports.
Audit Logging	Records alerts, recommendations, approvals, overrides, commands, and configuration changes.
Integration Monitoring	Tracks failed telemetry, weather, pricing, notification, identity, and automation integrations.

5. Candidate Data Stores

Data Store	Architectural Purpose
Configuration Store	Stores buildings, floors, zones, devices, equipment groups, thresholds, schedules, and integration settings.
Raw Telemetry Store	Stores received telemetry before or alongside normalization for troubleshooting and replay.
Normalized Time-Series Store	Stores normalized meter, sensor, environmental, and equipment readings.
Baseline Store	Stores expected usage baselines and reference profiles.
Alert Store	Stores alert records, severity, status, assignment, and resolution history.
Recommendation Store	Stores generated recommendations, rationale, approval state, and outcome.
Control Command Store	Stores command requests, target systems, acknowledgments, failures, and completion status.
Weather and Pricing Store	Stores weather observations, forecasts, utility pricing, and demand-response event data.

Data Store	Architectural Purpose
Occupant Feedback Store	Stores comfort feedback or environmental issue reports where enabled.
Reporting Store	Stores aggregates or derived metrics used for dashboards and reports.
Audit Log Store	Stores significant user, system, integration, and control events.

6. Candidate External Interfaces

Interface	Provider	Consumer	Notes
Telemetry Ingestion Interface	Smart Building Energy Management System	Meters, sensors, gateways, or automation systems	Receives raw readings and device status.
Building Automation Status Interface	Building Automation System	Smart Building Energy Management System	Provides equipment state, schedules, setpoints, and command feedback.
Building Automation Control Interface	Building Automation System	Smart Building Energy Management System	Executes approved control commands.
Utility Pricing Interface	Utility Pricing Feed	Smart Building Energy Management System	Provides tariff, time-of-use, peak demand, or demand-response data.
Weather Data Interface	Weather Data Provider	Smart Building Energy Management System	Provides weather observations and forecasts.
Notification Delivery Interface	Notification Provider	Smart Building Energy Management System	Sends alert and operational notifications.
User Authentication Interface	Identity Provider	Smart Building Energy Management System	Authenticates users and resolves roles or groups.
Occupant Feedback Interface	Smart Building Energy Management System	Occupant-facing experience	Captures comfort feedback and environmental issue reports.
Reporting Interface	Smart Building Energy Management	Facilities and analytics	Provides dashboard and reporting data.

Interface	Provider	Consumer	Notes
	System	experiences	

7. Primary Workflows for Architecture Modeling

Workflow	Summary
Ingest Telemetry	Receive, validate, normalize, store, and expose telemetry freshness.
Monitor Building Status	Display current energy, environmental, equipment, and alert state.
Detect Energy Anomaly	Compare current behavior to baseline or rules and generate alert.
Detect Sensor Health Issue	Identify stale, missing, invalid, or contradictory readings.
Generate Recommendation	Analyze telemetry, schedules, occupancy, weather, pricing, and constraints to propose action.
Approve Recommendation	Authorized user approves, rejects, dismisses, or modifies a proposed action.
Execute Control Command	Send approved command to building automation system and track acknowledgment or failure.
Override Control Action	Authorized user cancels or reverses an automated or approved action.
Handle Demand-Response Event	Ingest event, evaluate eligible actions, seek approval, execute, monitor, and summarize.
Send Alert Notification	Route high-priority alert through notification provider and record delivery status.
Generate Energy Report	Aggregate telemetry, baselines, recommendations, actions, and cost inputs into reports.
Capture Occupant Feedback	Receive comfort feedback and correlate with zone telemetry and recent actions.

8. Key State Models

Telemetry Reading State

Candidate states:

1. Received.
2. Validated.
3. Normalized.

4. Rejected.
5. Stale.
6. Missing.
7. Superseded.

Alert State

Candidate states:

1. New.
2. Notified.
3. Acknowledged.
4. Investigating.
5. Resolved.
6. Dismissed.
7. Escalated.

Recommendation State

Candidate states:

1. Generated.
2. Pending review.
3. Approved.
4. Rejected.
5. Expired.
6. Executed.
7. Failed.
8. Superseded.

Control Command State

Candidate states:

1. Pending.
2. Sent.
3. Acknowledged.
4. Rejected.
5. Failed.
6. Applied.
7. Reverted.
8. Timed out.

Demand-Response Event State

Candidate states:

1. Received.
2. Evaluating.
3. Pending approval.
4. Participating.
5. Completed.
6. Declined.
7. Failed.
8. Restored.

9. Key Quality Drivers

Quality Driver	Architectural Implication
Reliability	Telemetry, alerting, recommendations, and command tracking must survive temporary integration failures.
Safety	Control commands require strict guardrails and authorization boundaries.
Data Freshness	Dashboards and decisions must expose whether data is current, stale, missing, or invalid.
Auditability	Approvals, overrides, automated actions, command results, and configuration changes require durable audit records.
Scalability	Telemetry ingestion and time-series storage must support many devices across many buildings.
Observability	Operators must see integration failures, telemetry gaps, command failures, and alert backlogs.
Explainability	Recommendations should be traceable to readings, rules, baselines, pricing, weather, or schedules.
Interoperability	Device, meter, automation, pricing, weather, identity, and notification integrations require adapter boundaries.
Security	Control capability must be separated from read-only monitoring and restricted by role.
Privacy	Occupancy and feedback data should avoid unnecessary user identification.

10. Architecture Risks

Risk	Architectural Concern
Heterogeneous device protocols	Device and automation integrations may vary widely by building or vendor.
Stale telemetry driving bad decisions	Recommendation and control logic must account for data freshness and confidence.
Unsafe or uncomfortable control actions	Guardrails must prevent commands outside configured comfort or operating ranges.
Inconsistent command state	Commands may be sent, rejected, partially applied, or not acknowledged.
High telemetry volume	Time-series ingestion and storage may become a scaling bottleneck.
Baseline uncertainty	Early recommendations may be weak until enough historical data is available.
Over-alerting	Alert thresholds and anomaly logic must avoid operator fatigue.
Demand-response recovery	Temporary load reductions must be followed by reliable restoration behavior.
Integration failure visibility	External failures must be visible and actionable, not hidden in logs.
Occupant privacy	Feedback and occupancy data must be modeled carefully to avoid over-collection.

11. Open Architecture Questions

1. Which building automation systems and protocols must be supported first?
2. Are meters and sensors connected directly, through gateways, or through the building automation system?
3. What telemetry frequency is required for monitoring versus optimization?
4. Should raw telemetry be retained, or only normalized telemetry?
5. Which control actions are allowed in MVP, if any?
6. Which actions require human approval versus policy-based automation?
7. What guardrails are mandatory for temperature, humidity, ventilation, lighting, and equipment runtime?
8. Should recommendations expire if telemetry becomes stale?
9. What is the expected portfolio scale: buildings, zones, devices, and readings per minute?
10. How should demand-response participation be approved, monitored, and reversed?
11. What level of local operation is required if cloud connectivity is unavailable?

12. Should occupant feedback be anonymous, authenticated, or location-only?
13. What reports require near-real-time data versus historical aggregation?
14. Which failures require immediate alerting versus administrative review?

12. Recommended Modeling Scope

For the initial architecture model, focus on:

1. Telemetry ingestion and normalization.
2. Monitoring dashboard data flow.
3. Energy anomaly detection.
4. Recommendation generation and approval.
5. Notification delivery.
6. Time-series storage and reporting.
7. Audit logging.
8. Building automation status integration.

Defer advanced modeling for:

1. Automated control execution.
2. Demand-response orchestration.
3. Occupant comfort feedback.
4. Carbon reporting.
5. Predictive equipment analytics.
6. Multi-building benchmarking.